Potential energy

National 5 · Engineering contexts & challenges · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 Calculate the change in the rider's potential energy between the highest and lowest points.

- Top: 80 × 9.8 × 50 = 39 200 J
- Bottom: 80 × 9.8 × 5 = 3920 J
- Change = 35 280 J = 35 kJ

2 A 1.5 kg toy buggy rolls down ramps of different heights. Calculate its potential energy at the top of each ramp.

- 50 cm: 1.5 × 9.8 × 0.5 = 7.4 J
- 75 cm: 1.5 × 9.8 × 0.75 = 11 J
- 1 m: 1.5 × 9.8 × 1 = 15 J
- 1.24 m: 1.5 × 9.8 × 1.24 = 18 J

3 Using a ramp height of 1.22 m, calculate the potential energy of buggies of different mass.

- 2.5 kg: 2.5 × 9.8 × 1.22 = 30 J
- 3.5 kg: 3.5 × 9.8 × 1.22 = 42 J
- 6 kg: 6 × 9.8 × 1.22 = 72 J
- 400 g: 0.4 × 9.8 × 1.22 = 4.8 J

4a) Calculate the potential energy of 6 litres of water at the outlet pipe.

- m = 6 kg
- E_p = 6 × 9.8 × 8 = 470 J

4b) Describe why not all of this potential energy is converted into useful energy.

- Some energy is lost as heat and sound, due to friction between the water and the pipe.

5a) Calculate the potential energy of the ski jumper at point A.

- E_p = 65 × 9.8 × 80 = 50 960 J = 51 kJ

5b) At point B the ski jumper has 31.5 kJ of potential energy. Calculate the kinetic energy at B, assuming no energy is lost.

- $E_k = E_p(A) - E_p(B)$ = 50 960 − 31 500
- E_k = 19 460 J = 19 kJ

5c) Calculate the velocity of the ski jumper at point B.

- $v^2 = 2E_k \div m$ = 2 × 19 460 ÷ 65 = 598.8
- v = 24 m/s